Full Stack MERN Bookstore Application

Professional Project Report for GitHub and New Contributors

1. Project Overview

A production■level bookstore web application built using the MERN stack (MongoDB, Express.js, React.js, and Node.js). The system provides authentication, user dashboards, file uploads, and full CRUD operations for managing books. It follows modern full■stack architecture and demonstrates secure API development and responsive frontend design.

2. Key Features

• User authentication with JWT

• Secure password hashing using bcrypt

• User dashboard with profile management

• File uploads with Multer

• Protected routes and role■based access

• RESTful API architecture

• Modern UI using Tailwind CSS and DaisyUI

3. System Architecture

The application follows a client■server architecture where the React frontend communicates with the Express backend through REST APIs. MongoDB stores application data while JWT handles authentication and authorization.

4. Technology Stack

Layer	Technology
Frontend	React.js, React Router, Tailwind CSS, DaisyUI, Axios
Backend	Node.js, Express.js
Database	MongoDB with Mongoose
Authentication	JSON Web Tokens (JWT)
Security	bcrypt password hashing
File Upload	Multer middleware

5. Project Folder Structure

```
Bookstore/
■■■ Backend/
■   ■■■ controllers/
```

```
■   ■■■ middleware/
■   ■■■ models/
■   ■■■ routes/
■   ■■■ index.js
■■■ Front-end/
    ■■■ src/components
    ■■■ context
    ■■■ services
    ■■■ App.jsx
```

6. Authentication Flow

Users register or login through the frontend interface. The backend verifies the credentials and generates a JWT token which is returned to the client. The token is stored in the browser and attached to future API requests through the Authorization header.

7. File Upload Mechanism

The application allows users to upload profile pictures and documents. Files are sent as multipart/form■data requests. Multer middleware processes the upload, stores the file in the server uploads directory, and records metadata in MongoDB.

8. API Design

Method	Endpoint	Purpose
POST	/api/auth/signup	Register new user
POST	/api/auth/login	Authenticate user
GET	/api/books	Retrieve all books
POST	/api/books	Create new book
PUT	/api/books/:id	Update book
DELETE	/api/books/:id	Delete book

9. Security Implementation

• Passwords hashed with bcrypt

• JWT authentication with expiration

• Protected backend routes

• File type and size validation

• CORS protection

• Input validation on APIs

10. Running the Application

```
# Backend
cd Backend
npm install
```

```
npm run dev

# Frontend
cd Front-end
npm install
npm run dev
```

11. Conclusion

This project demonstrates a complete full■stack MERN application with secure authentication, modern frontend architecture, and scalable backend APIs. It serves as a strong foundation for production■level web applications and can be extended with features such as payment gateways, admin analytics dashboards, and book recommendation systems.